

Motion Synergy API 3rd-party Software Libraries

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1 Revision History

Issue	Date	By	Description
A	21 OCT 2021	Mark Gladding	Initial revision.
B	03 NOV 2021	Mark Gladding	Added NumericInput Control Library.
C	03 OCT 2023	Mark Gladding	Added OxyPlot Control Library.

2 Introduction

2.1 Purpose

This document details the 3rd-party Software Libraries and their associated licenses used by the MotionSynergyGUI and MotionSynergyAPI dll.

2.2 Scope

This document covers the following information related to 3rd-party libraries:

- Identification of 3rd-party libraries.
- Defines the functional and performance requirements for each 3rd-party library (covered by the *Intended Use* and *Selection Justification* sections for each library).
- Selection and validation strategy for each 3rd-party library.
- Details the license(s) for each 3rd-party library.
- Documents the title, version, manufacturer for each 3rd-party library.
- Assigns a unique designator for each 3rd-party library.
- Software (dependencies) necessary to support the proper operation of the 3rd-party library.

2.3 Audience

This document provides a reference of selected 3rd-party libraries and a record of the selection criteria used for each 3rd-party library.

The intended audience for this document includes:

- Software team members charged with the design and implementation of the software.
- Software and system testers charged with verifying the correct operation of the software.
- Future software developers charged with maintenance of the software.

3 3rd-party Software Libraries

3.1 Boost C++ Library

ID IMP-SOUP-BOOST

Title Boost C++ Library

Version 1.89.0

Package Name(s) boost-atomic boost-chrono boost-crc boost-date-time boost-filesystem boost-format boost-json boost-log boost-multi-array boost-program-options boost-regex boost-system boost-thread boost-uuid

Manufacturer The Boost Foundation

Description Boost is a set of high-quality, free, peer-reviewed, open-source cross-platform C++ libraries. It works well with the C++ Standard Library. Use of Boost reduces initial development costs, results in fewer bugs, reduces reinvention-of-the-wheel, and cuts long-term maintenance costs.

URL <https://www.boost.org/>

Intended Use Boost has two primary purposes:

1. Provide an abstraction layer which hides operating system specifics. This allows cross-platform (e.g. Linux and Windows) applications to be developed with minimal effort.
2. Provides a set of general purpose libraries which reduce the amount of code needed to develop a specific application. Some commonly used libraries include asio (network and serial communications, timers), regex (regular expressions), property trees (configuration file support), etc.

Selection Justification Following is the list of attributes used in the selection of this library:

- It is a mature library with wide adoption across the software industry.
- It has active development and patches.
- Royalty-free license conditions permitting use in a closed source, commercial product.
- Cross-platform support for both Linux and Windows.
- Minimal or no dependencies on other libraries.
- Functions correctly within the CPU, memory and persistent storage resource constraints of the Central Controller software runtime environment.

License BSL-1.0

Limitations No known issues impacting Dependent Units.

Validation Strategy Validated as part of the unit/integration tests of dependent Software Units in the required context.

3.2 C++ Standard Library (Windows)

ID

IMP-SOUP-CPPLIBWIN

Title

C++ Standard Library (Windows)

Version

Microsoft Visual Studio 2022 v17.7

Microsoft Visual C++ Toolset v143

Manufacturer

Microsoft

Description

Provides an implementation of the standard libraries associated with the C++ language. These libraries include:

- C++ Standard Library
- C Standard Library
- C Maths Library

URL<https://visualstudio.microsoft.com/>**Intended Use**

The C++ standard library has two primary purposes:

1. Provide an abstraction layer which hides operating system specifics. This allows cross-platform (e.g. Linux and Windows) applications to be developed with minimal effort.
2. Provides a set of general purpose libraries which reduce the amount of code needed to develop a specific application. Some commonly used libraries include containers, shared pointers, threading, etc.

Selection Justification

Included as part of the Microsoft Visual Studio C++ toolset, a high quality, continuously maintained product used industry-wide for development on the Microsoft Windows platform.

License[Microsoft Visual Studio License](#)**Dependent Units****Dependencies**

None.

Limitations

No known issues impacting Dependent Units.

Validation Strategy

Validated as part of the unit/integration tests of dependent Software Units in the required context.

3.3 Lua Scripting Library

ID

IMP-SOUP-LUA

Title

Lua Scripting Library

Version

5.4.8

Package Name(s)

lua

luainterpreter

Manufacturer

PUC-Rio

Description

Lua is a powerful, efficient, lightweight, embeddable scripting language. Lua is dynamically typed, runs by interpreting bytecode with a register-based virtual machine, and has automatic memory management with incremental garbage collection, making it ideal for configuration, scripting, and rapid prototyping. Several benchmarks show Lua as the fastest language in the realm of interpreted scripting languages. Lua is fast not only in fine-tuned benchmark programs, but in real life too.

URL<https://www.lua.org/>**Intended Use**

Lua provides a scripting language for creating workflows to coordinate the control of multiple hardware devices. Typical workflows include initialising hardware, performing a power on self test, running an assay, performing a maintenance function such as emptying waste, etc. Lua scripting is not intended to perform hard realtime operations (i.e. requiring sub-50ms timing precision) which involve coordination of two or more hardware devices due to latency and jitter between the script command execution and the hardware control.

Selection Justification

Lua has been used in many industrial applications (e.g., Adobe's Photoshop Lightroom), with an emphasis on embedded systems (e.g., the Ginga middleware for digital TV in Brazil) and games (e.g., World of Warcraft and Angry Birds). Lua is currently the leading scripting language in games. Lua has a solid reference manual and there are several books about it. Several versions of Lua have been released and used in real applications since its creation in 1993.

Lua was selected as it can be run on both the Central Controller and the resource-constrained embedded environment of the Subsystem Controller. Currently Lua is only used on the Central Controller.

License

MIT

Copyright Copyright © 1994–2024 Lua.org, PUC-Rio.

Limitations

No known issues impacting Dependent Units.

Validation Strategy

Validated as part of the unit/integration tests of dependent Software Units in the required context.

Modifications***Lua Library***

The lua library is built using the Lua vcpkg and has the 'cpp' feature specified to build and compile the library as C++ rather than the default C library. This library is linked with the Central Controller software to allow execution of Lua scripts.

Lua Interpreter

A version of the lua interpreter (lua.exe) is built which includes a statically linked luafilesystem library. This executable is only used to generate Lua documentation using LDoc and Penlight. A local *luainterpreter* port has been created to achieve this. It does this as follows:

1. Downloads and extracts the luafilesystem source archive from github.
2. Applies a patch to lfs.c to resolve some compiler warnings
3. Downloads and extracts the lua source archive from lua.org.
4. Applies a patch to open the lfs library during the openlibs call at startup.
5. Copies the lfs source code into the lua source folder.
6. Builds lua.exe using the lua and lfs source code.

Details can be found in [luainterpreter/5.4.6_0/portfile.cmake](#)

3.4 MathNet.Numerics - Maths Library for .NET

ID

IMP-SOUP-MATHNET

Title

MathNet.Numerics - Maths Library for .NET

Version

5.0.0

Manufacturer

Christoph Ruegg, Marcus Cuda, Jurgen Van Gael

Description

Math.NET Numerics is the numerical foundation of the Math.NET project, aiming to provide methods and algorithms for numerical computations in science, engineering and every day use.

URL

<https://numerics.mathdotnet.com/>

Intended Use

Provides linear algebra and FFT capabilities in the MotionSynergyGUI.

Selection Justification

Following is the list of attributes used in the selection of this library:

- It is a mature library with wide adoption across the software industry.
- It has active development and patches.
- Royalty-free license conditions permitting use in a closed source, commercial product.
- Cross-platform support for Linux and Windows.
- Minimal or no dependencies on other libraries.
- Functions correctly within the CPU, memory and persistent storage resource constraints of the MotionSynergyGUI software runtime environment.

License

[MIT License](#)

Dependent Units

- MotionSynergyGUI

Dependencies

- System.ValueTuple

Limitations

No known issues impacting Dependent Units.

Validation Strategy

Validated as part of the unit/integration tests of dependent Software Units in the required context.

3.5 NumericInput Control Library

ID

IMP-SOUP-NUMERIC-INPUT

Title

NumericInput Control Library

Version

0.5.5

Manufacturer

Johan Larsson

Description

WPF TextBox for numeric input. Includes controls DoubleBox, IntBox, DecimalBox, FloatBox and ShortBox.

URL

<https://github.com/GuOrg/Gu.Wpf.NumericInput>

Intended Use

Provides controls for numeric input via the MotionSynergyGUI.

Selection Justification

Following is the list of attributes used in the selection of this library:

- It is a mature library with wide adoption across the software industry.
- It has active development and patches.
- Royalty-free license conditions permitting use in a closed source, commercial product.
- Cross-platform support for Linux, QNX and Windows.
- Minimal or no dependencies on other libraries.
- Functions correctly within the CPU, memory and persistent storage resource constraints of the MotionSynergyGUI software runtime environment.

License

[MIT License](#)

Dependent Units

- MotionSynergyGUI

Dependencies

None

Limitations

No known issues impacting Dependent Units.

Validation Strategy

Validated as part of the unit/integration tests of dependent Software Units in the required context.

3.6 OxyPlot - Plotting Library for .NET

ID

IMP-SOUP-OXYPLOT

Title

OxyPlot - Plotting Library for .NET

Version

2.2.0

Manufacturer

OxyPlot contributors

Description

OxyPlot is a cross-platform plotting library for .NET. A number of support packages are available to support a specific UI framework.

URL

<https://oxyplot.github.io/>

Intended Use

Provides plotting capabilities in the MotionSynergyGUI.

Selection Justification

Following is the list of attributes used in the selection of this library:

- It is a mature library with wide adoption across the software industry.
- It has active development and patches.
- Royalty-free license conditions permitting use in a closed source, commercial product.
- Cross-platform support for Linux and Windows.
- Minimal or no dependencies on other libraries.
- Functions correctly within the CPU, memory and persistent storage resource constraints of the MotionSynergyGUI software runtime environment.

License

[MIT License](#)

Dependent Units

- MotionSynergyGUI

Dependencies OxyPlot.Core package implements the core plotting related APIs. However, to use OxyPlot with a UI framework, support packages for the UI framework are required. For example, OxyPlot.Wpf and OxyPlot.Wpf.Shared packages are required for OxyPlot to work with WPF UI framework.

Limitations

No known issues impacting Dependent Units.

Validation Strategy

Validated as part of the unit/integration tests of dependent Software Units in the required context.

3.7 WPFCustomMessageBox - Customizable WPF MessageBox

ID

IMP-SOUP-WPFCUSTOMMESSAGEBOX

Title

WPFCustomMessageBox - Customizable WPF MessageBox

Version

1.0.7

Manufacturer

Evan Wondrasek / Apricity Software LLC

Description

A WPF clone of the native Windows/.NET MessageBox with extra features like custom button text.

URL

<https://github.com/evanwon/WPFCustomMessageBox>

Intended Use

Provides a more customizable WPF message box e.g., customizable button texts.

Selection Justification

Following is the list of attributes used in the selection of this library:

- It is a mature library with wide adoption across the software industry.
- It has active development and patches.
- Royalty-free license conditions permitting use in a closed source, commercial product.
- Minimal or no dependencies on other libraries.
- Functions correctly within the CPU, memory and persistent storage resource constraints of the MotionSynergyGUI software runtime environment.

License

[MIT License](#)

Dependent Units

- MotionSynergyGUI

Dependencies None

Limitations

No known issues impacting Dependent Units.

Validation Strategy

Validated as part of the unit/integration tests of dependent Software Units in the required context.

4 License Text

4.1 Motion Synergy API License Text

4.1.1 Boost C++ Library License Text

Boost Software License - Version 1.0 - August 17th, 2003

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4.1.2 C++ Standard Library (Windows) License Text

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MICROSOFT VISUAL C++ 2015 - 2022 RUNTIME

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EULA ID: Cpp_2015-2022_ENU.1033

4.1.3 Lua Scripting Library License Text

Copyright © 1994–2025 Lua.org, PUC-Rio.

<Standard MIT License Text>

4.1.4 MathNet.Numerics - Maths Library for .NET License Text

MIT License

Copyright (c) 2002-2022 Math.NET

<Standard MIT License Text>

4.1.5 NumericInput Control Library License Text

The MIT License (MIT)

Copyright (c) 2015 Johan Larsson

<Standard MIT License Text>

4.1.6 OxyPlot - Plotting Library for .NET License Text

MIT License

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<Standard MIT License Text>

4.1.7 WPFCustomMessageBox - Customizable WPF MessageBox License Text

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4.2 Standard License Text

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